

## COLDPOD · REQUIREMENTS

# ColdPod requirements

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## DOCUMENT CONTROL

|             |                                       |         |                |
|-------------|---------------------------------------|---------|----------------|
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| Author      | Amish Chadha                          | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept         |         |                |
| Repository  | github.com/BoujeeEnjinia1701/coldpod  |         |                |
| Source      | docs/03-requirements.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                   |
|---------|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.8     | 2026-10-02 | Amish Chadha | Decisions of 2026-10-02: R8 restated as 9 h, R12 relaxed to 5.75 kg for the prototype, R14 removal method for the prototype; R8 and R12 now met on paper |
| 0.7     | 2026-10-01 | Amish Chadha | Budget treated as a value-engineering target                                                                                                             |
| 0.6     | 2026-10-01 | Amish Chadha | Status values from CPD-CAL-001 v0.4 for the constructable design (CPD-DDR-003, Draft); no target changed                                                 |
| 0.5     | 2026-09-26 | Amish Chadha | Budget top-up approved by Amish; R15 target \$310, status met against CPD-CAL-001 v0.3                                                                   |
| 0.4     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002). R2 and R7 targets restated for the liner cut-out and buck-boost driver; status from CPD-CAL-001 v0.2        |
| 0.3     | 2026-09-25 | Amish Chadha | Record Amish's decisions (CPD-DDR-001); R6 redefined to 16 h, R12 relaxed to 5.5 kg, R15 set to \$300; status from CPD-CAL-001                           |

|     |            |              |                                                                                    |
|-----|------------|--------------|------------------------------------------------------------------------------------|
| 0.2 | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with status against the concept estimates |
| 0.1 | 2026-09-24 | Amish Chadha | Initial scaffold                                                                   |

# ColdPod requirements

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These are the requirements for the concept, checked by calculation at TRL 3 in CPD-CAL-001 v0.4 against the constructable design of CPD-DDR-003, which Amish accepted on 2026-10-02. Amish decided the TRL 2 review items on 2026-09-25 (CPD-DDR-001): R6 is redefined from 24 h to 16 h at 43 °C (D5), R12 is relaxed from 5.0 kg to 5.5 kg (D6), R15 follows the \$300 value-engineering target of D10 and the R10 alarm defaults are adopted (D9). He then accepted the TRL 3 recommendations (CPD-DDR-002): R2 now names the two hardware cut-outs in series and R7 names the buck-boost driver that serves the full 10 to 15 V vehicle range; no target was relaxed. The target is now \$310 (CPD-DDR-002 N3), a hypothetical control target and not a limit, and R15 follows it. No target changed in v0.6; the values follow the constructable design. Against CPD-CAL-001 v0.4, ten requirements were met, three at risk, two **not met** by small margins (R8 and R12) and two not verifiable at TRL 3 (R10 and R16). On 2026-10-02 Amish relaxed R12 to 5.75 kg for the prototype and restated R8 as 9 h (CPD-DEC-001), so twelve are now met on paper, three at risk and two not verifiable; R14's removal method was decided for the prototype. Before DDR-002 the count was seven met, five at risk and three not met (R2, R8, R12).

Table 1. Requirements.

| ID  | Requirement                                    | Target                                                                                                                                                                                                                                                                                                                                        | Verification (TRL 3 or later)                                  | Status at TRL 3 (CPD-CAL-001)                                                                                                            |
|-----|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Keep the payload in the safe range             | 2 to 8 °C at a buffered probe anywhere in the payload space; control setpoint 5 °C                                                                                                                                                                                                                                                            | Heat-transfer calculation; later lab test in a climate chamber | Met on paper: liner held at 2 to 5 °C; the lid pack stays frozen near 2.8 °C on its cold plate in powered hold at 43 °C (was at risk)    |
| R2  | Never freeze the payload                       | No payload-contact surface below 1 °C in normal operation or with a stuck-on Peltier driver; two hardware cut-outs in series in the Peltier supply, on the cold block (−5 °C) and on the liner (3 °C), independent of firmware (CPD-DDR-002)                                                                                                  | Calculation of the cold path; fault analysis                   | Met on paper: with a stuck-on driver the liner settles near 2.0 °C after the liner cut-out opens (was not met, about −4 °C)              |
| R3  | Carry a useful load                            | 1.0 L or more of usable payload; holds 20 or more insulin pens up to 170 mm long                                                                                                                                                                                                                                                              | Massing model                                                  | Met: 1.37 L, 24 pens, up to 172 mm                                                                                                       |
| R4  | Hold with no power at all                      | 12 h or more at a constant +43 °C, starting with the PCM frozen                                                                                                                                                                                                                                                                               | Heat-leak and PCM calculation                                  | At risk: 13.3 h; 10.2 h with 30 % more heat leak                                                                                         |
| R5  | Hold off-grid on a full charge                 | 24 h or more at a constant +32 °C on the internal battery, then the PCM                                                                                                                                                                                                                                                                       | Energy budget                                                  | At risk: 27.3 h; 21.4 h with 30 % more heat leak                                                                                         |
| R6  | Hold off-grid on a full charge in extreme heat | 16 h or more at a constant +43 °C on the internal battery, then the PCM; longer hot trips run from external power under R7 (redefined from 24 h, CPD-DDR-001 D5)                                                                                                                                                                              | Energy budget                                                  | At risk: 16.7 h; 12.6 h with 30 % more heat leak                                                                                         |
| R7  | Hold indefinitely on external power            | Keeps 2 to 8 °C up to +43 °C from 12 V DC (10 to 15 V, through a buck-boost Peltier driver, CPD-DDR-002) or USB-C PD (20 V, 45 W or more)                                                                                                                                                                                                     | Peltier and heat sink calculation                              | Met on paper: 18.8 W input, well inside 45 W; the buck-boost stage covers a 10 V input; the lid pack stays frozen (was at risk)          |
| R8  | Recharge the cold quickly                      | Refreeze a fully melted PCM in 9 h or less at 25 °C ambient, box empty, while charging the battery (restated from 8 h, CPD-DEC-001, 2026-10-02; still an overnight refreeze)                                                                                                                                                                  | PCM and Peltier calculation                                    | Met: 8.0 h (8.03 h) for all the PCM (jacket 6.8 h, lid pack 8.0 h); 8.4 h before the constructable design, which holds slightly less PCM |
| R9  | Record the temperature                         | Payload probe accuracy $\pm 0.5$ °C; log every 1 min; 60 days or more on board; export as CSV over Bluetooth Low Energy or USB                                                                                                                                                                                                                | Datasheet and storage calculation; later calibration           | Met: 1.38 MB of 2.10 MB; accuracy by sensor selection                                                                                    |
| R10 | Raise alarms early                             | Local sound, light and display alarm, plus a phone notification when paired: warn after 10 min outside 2 to 8 °C; alarm at once at 0 °C or lower on the payload probe; early freeze warning at 1 °C on the liner probe; low battery, sensor fault and lid open for more than 2 min (defaults decided, CPD-DDR-001 D9; adjustable per product) | Design review of firmware sketch                               | Not verifiable at TRL 3: design intent only, no firmware sketch                                                                          |
| R11 |                                                |                                                                                                                                                                                                                                                                                                                                               | Power budget                                                   | Met: about 96 days                                                                                                                       |

| ID  | Requirement                         | Target                                                                                                                                          | Verification (TRL 3 or later)    | Status at TRL 3 (CPD-CAL-001)                                                                                                                                                                                                                             |
|-----|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | Keep logging when the cooling stops | Logger runs 14 days or more after the Peltier is shut off for low battery                                                                       |                                  |                                                                                                                                                                                                                                                           |
| R12 | Be light enough to carry all day    | 5.75 kg (12.7 lb) or less empty for the prototype (relaxed from 5.0 kg, CPD-DDR-001 D6, and from 5.5 kg, CPD-DDR-003 A2, 2026-10-02)            | Mass estimate, weighing at TRL 4 | Met on paper: 5.70 kg, 0.05 kg margin (5.53 kg before the fixings and parts added to make the design buildable, CPD-DDR-003); no base bumper until the prototype is weighed                                                                               |
| R13 | Be compact                          | Fits in 400 x 250 x 250 mm including handle                                                                                                     | Massing model                    | Met: 366 x 220 x 207 mm                                                                                                                                                                                                                                   |
| R14 | Travel by air                       | Battery 100 Wh or less, removable or with a shipping switch                                                                                     | Battery specification            | Met for the prototype: 76.8 Wh; the pack comes out with the battery bay (four screws), counted as removable for the prototype (CPD-DDR-003 A3, 2026-10-02). Before any field or air travel use: a shipping switch and the carrier's lithium battery rules |
| R15 | Be affordable and buildable         | Estimated parts cost within the \$310 value-engineering target (hypothetical control target, CPD-DDR-002 N3); no custom PCB for the first build | Priced BOM                       | Within the value-engineering target: \$308 against \$310, with the construction parts of CPD-DDR-003                                                                                                                                                      |
| R16 | Survive field use                   | Splash resistant (IP54 target for electronics bays); survives a 0.5 m drop onto a hard floor while loaded                                       | Design review; later test        | Not verifiable at TRL 3                                                                                                                                                                                                                                   |
| R17 | Be clearly labelled as a prototype  | "Research prototype, not a medical device" on the box, the display start screen and every exported log                                          | Design review                    | Met by design                                                                                                                                                                                                                                             |

## ■ Assumptions

- Ambient cases follow the WHO PQS test point of +43 °C for the hot zone ([WHO PQS E004/VC01](#)), +32 °C for a typical hot day and 25 °C for a clinic.
- The 2 to 8 °C window and the no-freezing rule come from insulin and vaccine storage guidance ([US FDA](#); [CDC Pink Book, chapter 5](#)).
- Logging and alarm targets are stricter than a WHO 30-day recorder (−0.5 °C for 60 min, +8 °C for 10 h; [TechNet-21](#)) because a carrier in transit can move out of range much faster than a refrigerator.
- R6 matches the WHO long-range carrier cold life only loosely: a WHO carrier is tested passive for 30 h at 43 °C. ColdPod does not claim WHO equivalence.
- The alarm thresholds in R10 were decided by Amish on 2026-09-25 (CPD-DDR-001 D9) and will be adjustable per product.

- Hold-time status uses a  $\pm 30\%$  band on the heat leak: met if the target holds at  $+30\%$ , at risk if it holds only at the nominal estimate (CPD-CAL-001).
- R15 counts ColdPod parts only; the payload is not costed.